

IMMUNOELECTROADSORPTION THE FUNDAMENTALS OF THE IMMUNOELECTROADSORPTION METHOD*

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Abstract—The fundamentals involved in the newly developed immunoelectroadsorption method to detect immunological reactions are described. An immunoelectroadsorption assay consists in adsorbing on a metalized slide first a layer of antigen and then a layer of antibodies. Both adsorptions are carried out with the help of a small electric current with the appropriate polarity. The thickness of the layer adsorbed from the antiserum is a function of the concentration of antibodies. The conditions under which the assay is to be conducted are described and the results obtained with the antigens serum bovine albumin, ovalbumin, and growth hormones are reported. These results show that the method is extremely sensitive, concentrations of growth hormone as small as 10^{-9} g/ml can be estimated even in the presence of a large excess of foreign protein.

In the past few years a new method called Immunoelectroadsorption (IEA) has been developed in this laboratory to detect circulating antibodies in the course of viral [1–4] and parasitic infections [5–7]. This method was also found useful for the quantitative measurement of the amount of circulating growth hormone in the physiological range (10^{-10} – 10^{-8} g/ml) [8]. It appears important at this time to review the basic principles involved since other laboratories are investigating this new method as a diagnostic tool.

The method has its origin in the observation made years ago [9] that monolayers of antigenic proteins deposited on metalized glass slides were capable of adsorbing specifically antibodies. The adsorption was measured by optical means.

Thus a very simple method, which had possibilities for wide applications, was available for the detection of immunological reactions. It proved, however, to have serious limitations because the antigen solutions had to be relatively pure and concentrated, and the antisera could not be greatly diluted.

However, when an electric current with the proper polarity was applied during the antigen and antibody adsorption periods, the slide upon which the deposition took place being one of the electrodes, clearcut differentiation between specific and non-specific adsorption could be obtained in spite of the fact that the antigenic preparations were very crude [1]. It became clear that the role of the current was (1) to maintain by electrophoresis a larger concentration of antigen as well as of antibodies in the immediate vicinity of the surface properly charged and (2) to prevent or reduce the adsorption of material of the same sign as that of the slide.

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EXPERIMENTAL

The test consists of two successive electroadsorptions on a slide. The antigen is deposited in the first and the antibodies in the second adsorption; the adsorptions are carried out as follows: a glass slide ($12\text{ cm} \times 0.6\text{ cm} \times 0.15\text{ cm}$), coated by evaporation with a metallic layer roughly $4000\text{ \AA}$ thick, is introduced into a test tube 0.8 cm in diameter filled with 0.5 ml of the antigen solution. Depending on the antigen and the pH of the solution, the slide is connected to either the positive or negative pole of a constant current power supply and a thin platinum wire 0.075 cm sec dia. serves as the other electrode. The adsorption period lasts 30 sec or more and the current intensity is kept constant, between 100 and $300\text{ }\mu\text{A}$. After washing and drying with a hand blower, the thickness of the adsorbed layer is measured with an ellipsometer within $\pm 0.2\text{ \AA}$. For the second adsorption 0.4 ml of an immune serum, diluted at least $1/10$ and up to $1/10,000$ in veronal buffer 0.03 M pH 7.7 , is introduced in the test tube. The slide is then always connected to the positive pole; the current intensity is in general the same as that used in the first adsorption and the adsorption period varies between 30 sec and 2 min . After washing and drying of the slide the optical thickness of the layer adsorbed from the serum is again determined with an ellipsometer. To obtain comparable results it is important to keep the sizes of slides and test tubes the same and to use a constant volume of solutions. Otherwise variations in the thickness adsorbed could result from changes in the current density if the immersed area of the slides is not kept constant. One single slide can be used for five successive experiments if its lower part, about 1.2 cm long, which has been immersed is cut off before using the slide for another experiment.

Metalization of the slides by evaporation

Of all the metals tested (chromium, nichrome, Inconel, rhodium and titanium) chromium appears the most satisfactory. One of the greatest difficulties encountered in the reproducibility of the IEA tests appears to reside in the properties of the metallic surface. Certain batches of slides supposedly metalized under the same conditions yielded poor or negative results, that is the specific adsorption could not be differentiated from the non-specific one. To obviate this difficulty, the metalized slides were coated before use with monolayers of stearate layers as described in the next section.

A metallic oxide film is formed during the electro-deposition when the slides are positively charged. For instance, under similar conditions ($300\text{ }\mu\text{A}$, veronal buffer 0.03 M pH 7.7 for 1 min) an increase in optical thickness of $100\text{ \AA}$ is observed on titanium, $35\text{ \AA}$ on Inconel and only $9\text{ \AA}$ on chromium coated slides. This is one of the reasons for choosing chromium for coating the slides. The metallic oxide film is detrimental to the accurate determination of the thickness of the protein layers.

Coating of the metalized slides with Ba stearate monolayers

Chromium plated slides coated with five monolayers of Ba stearate were found very consistent in their performance in an IEA test. The deposition of the layers was made according to the method of Blodgett and Langmuir[10]. The metalized slides, after cleaning with a watery paste of cerium oxide, were rinsed

and introduced still wet into a Langmuir trough before the formation of the surface film of Ba stearate. The film was then spread on the surface and the first monolayer was transferred during the emersion ($\uparrow$) of the slide, the other four layers being deposited by successive additional immersions and emersions ($\downarrow$)₂.

Polarity of the slides for the electrodepositions of antigens and antibodies

A clean metallic surface, prior to any deposition of layers, has a strong affinity for protein molecules which can therefore be adsorbed whatever the charge of the slide as long as the pH of the solution is not too far removed from the isoelectric point of the protein. This is the reason why the polarity of the slide plays a secondary role during the adsorption of the antigen, especially so if the concentration of the material is relatively high and the solution poorly buffered. It did not matter whether a slide during the adsorption of a 1 per cent bovine albumin solution in distilled water was neutral, positively or negatively charged, the layers subsequently adsorbed from an immune rabbit serum and a normal rabbit serum were 100 Å and 50 Å thick respectively. When the pH of the bovine albumin solution was kept at pH 3.3 no adsorption of albumin took place with the slide positively charged, since the same increment in thickness of 74 Å was observed after the subsequent adsorption from either an immune or a normal rabbit serum. However, if the slide was negatively charged during the adsorption of the albumin, 88 Å and 59 Å could be adsorbed from an immune and normal serum respectively. For the second adsorption, namely that of the antiserum, the slides were always positively charged since the solution was strongly buffered at pH 7.7, well above the isoelectric point of the antibodies. When the slides were negatively charged, the thickness adsorbed from both immune and normal sera was the same and very much smaller ≈ 10 –13 Å.

Optimal conditions for an IEA test

The difference in the thickness adsorbed from an immune serum and that adsorbed in a control experiment should be as great as possible. The ratio of the thicknesses obtained in one experiment and its control is a better measure of the immunological reaction than the difference of the thicknesses [6]. Furthermore, the accuracy with which a thickness can be determined decreases with the increase in thickness. A film a few Å to 80 Å thick can be easily determined within ± 0.02 Å, whereas the same accuracy cannot be obtained if the film is a few hundreds Å thick.

Control experiments

It should be emphasized that the layer adsorbed from an immune serum on a slide upon which a layer of antigen has been deposited does not consist exclusively of antibodies, but that an appreciable non-specific adsorption takes place. At first sight it would appear that an adequate control would consist in adsorbing a layer from a normal serum on another part of the slide coated by the same antigen, the difference in thickness in the two cases being a measure of the immunological reaction. This is very often a sufficient control. However, in the case of human sera it entails many pitfalls because it may happen that

some of the heterologous sera used as controls contain larger amounts of globulins than the homologous serum and consequently the non-specific thickness adsorbed could be larger than the thickness obtained with the homologous serum. Hence the thickness *per se* adsorbed from a serum is not always indicative of an immunological reaction. The following test permits a decision as to whether an appreciable difference in the thickness adsorbed from a homologous and heterologous serum corresponds to an immunological reaction. Instead of using two different sera on two slides coated with the same antigen, the same immune serum is adsorbed on two slides coated either with a carrier solution containing different amounts of the antigen or with the carrier alone. For instance, in the case of viral infections, the agent (antigen) is obtained from crudely purified extracts of organs of diseased animals (spleens of mice for the Friend virus, brains of newborn mice for arthropodborne viruses). The antigenic material is but a small fraction of the preparation. Hence a good control is made by adsorbing on a slide an extract obtained from a normal organ prepared by the same procedure followed for the extraction of the diseased organ. The occurrence of an immunological reaction is indicated by the difference in thickness of the layer adsorbed from the same immune serum on two slides, one coated with the extract of the diseased organ and the other with the extract of the normal organ. If the immune antiserum is heterologous, no difference in the thickness adsorbed is observed between the two slides.

When minute amounts of pure antigens are to be determined by the IEA technique, it is important to dissolve the antigen in a solution which contains some foreign proteins which can be in relatively large concentrations as compared to the antigen. These solutions used for dissolving the antigen are called carriers.

RESULTS

Series of experiments were carried out with pure antigens, i.e. bovine serum albumin, ovalbumin, human and bovine growth hormones and their corresponding rabbit antisera for the main purpose of testing the range of applicability of the IEA assay.

Three double layers of bovine albumin formed on a water surface in a Langmuir trough were transferred to nichrome plated slides. Without a current, the thickness adsorbed from a homologous antiserum was always greater than that adsorbed from a heterologous one as long as the dilutions of the antiserum in veronal buffer 0.03 M pH 7.7 were not greater than 1-500. At that dilution of 1/500 the adsorbed layer was only a few Å thick. However, when the slides were positively charged during the adsorption carried out to 3 min with a current of 300 µA, 34 Å were adsorbed from an antiovine albumin and 17 Å from an antiovalbumin rabbit serum diluted 1/500. At a 1-3000 dilution under the same conditions, 23 Å and 2 Å were adsorbed from the antiovine and antiovalbumin sera respectively.

When two double layers of bovine albumin were transferred to a slide coated with three monolayers of stearate, one could still observe an immunological reaction at dilutions of 1/10,000 of the antiserum (50 Å from homologous and 30 Å from heterologous serum in 5 min with 300 µA). These results show that if a current is used for the adsorption of an antiserum on a slide coated with

a pure antigen, and immunological reaction can be detected at dilutions far greater than in the absence of a current.

Determination of small concentrations of serum bovine albumin and ovalbumin in aqueous solutions with corresponding antisera

The electroadsorptions were carried out directly on Inconel slides negatively charged for the adsorption of the antigen. The albumins were dissolved in water and the antisera diluted 1/20 in veronal buffer pH 7.7. Had the albumins been dissolved in veronal pH 7.7, no specific adsorption from a homologous serum could have been detected. The adsorption times for the antigen and antibodies were one and two minutes respectively with a current of 300 μ A. The results are summarized in Table 1. The same antisera were used for all the experiments.

Table 1. Immunoelctroadsorption of antisera on slides coated with pure ovalbumin or pure bovine albumin

Conc. bovine albumin γ /ml in water	Thickness in Å from antibovine albumin serum	Thickness in Å from anti- ovalbumin serum	Conc. ovalbumin γ /ml in water	Thickness in Å from antibovine albumin serum	Thickness in Å from anti- ovalbumin serum
10	97	43	100	61	88
0.08	96	43	10	66	67
0.04	95	62			
0.01	95	74			
0.0	$\approx$ 100	80			

It should be noted that the thickness adsorbed from the antibovine albumin serum was independent of the concentration in bovine albumin of the solution used for the first electroadsorption (from 10 γ to 0 γ /ml), whereas the layer adsorbed from the heterologous antiovalbumin serum, kept increasing from 43 Å to 80 Å in the same range of concentrations of the antigen. This increase in the non-specific adsorption comes from the fact that the thickness of the layer of any protein when electroadsorbed on a clean metallic slide is always larger than that observed when there is already a protein layer on the slide before the adsorption. Therefore, when solutions of low concentrations in bovine albumin were used in the first adsorption, the metallic surface became sparsely covered and the thickness of non-specific adsorption was larger than that observed when the metallic surface had been fully covered in the first adsorption where more concentrated solutions of bovine albumin had been used. This shows that the non-specific adsorption from a serum depends a great deal on the degree of saturation of the surface prior to adsorption. As seen from Table 1 at zero concentration of bovine albumin, the nonspecific adsorption amounted to 100 Å for the antibovine albumin serum and 80 Å for the antiovalbumin. The antiovalbumin serum was weak but, as it appears clearly from the Table, the thickness of the layer adsorbed from an antibovine albumin serum was larger when the slide was coated with bovine albumin rather than with ovalbumin, and inversely

the thickness from the antiovalbumin was larger on a slide coated with ovalbumin rather than with bovine albumin.

Determination of small concentrations of serum bovine albumin in 2 per cent normal rabbit serum solutions

As mentioned earlier, the antigenic materials are generally crude preparations obtained from organ extracts, or they may constitute but a very small part of the fluid to be tested (for instance growth hormone in serum). Therefore, it was essential to investigate the influence of a large excess of foreign material in the solution of the antigen on the subsequent specific adsorption from the immune serum. To this effect, solutions of serum bovine albumin were prepared in concentrations ranging from 1 γ to 0.05 γ per ml in a 2 per cent normal rabbit serum (water was used for the dilution of the serum). Thus the ratio of antigen to foreign proteins varied between $\approx 10^{-4}$ to $\approx 10^{-6}$. The tests were performed as described above. A 2 per cent normal rabbit serum solution was used as a carrier for the control and the same immune serum served for all the experiments except those reported in row one and two of Table 2 which summarizes the results. The adsorptions were carried out directly on Inconel plated slides. The total adsorption time for the immune serum lasted 6 min, but the slides were washed, dried and read after two and four minutes. On account of fluctuations in the thickness adsorbed from the immune serum resulting from variations in the characteristics of the metallic surface, control experiments were performed

Table 2. Immunoelctroadsorption of antisera on slides coated with protein and sparsely populated with bovine albumin

Conc. bovine albumin in 2% normal rabbit serum (γ /ml)	Thickness in Å units of adsorbed layers from immune serum after					
	2 min		4 min		6 min	
		$\Delta\text{Å}$		$\Delta\text{Å}$		$\Delta\text{Å}$
{ 200	100	35				
{ 0	65					
{ 1.0 in 5% rabbit 0.0 serum	52	15	83	33		
	37		50			
{ 0.5	61	10	125	26	165	35
	0.0		99		130	
{ 0.17	66	21	125	28	180	48
	0.0		97		132	
{ 0.17	61	14	117	25	180	33
	0.0		82		147	
{ 0.05	60	6	113	8	160	
	0.0		105		156	

on the same slide after each experiment with bovine albumin solutions. Experiments and corresponding controls have been bracketed in Table 2. It is apparent that specific adsorption could still be detected when the concentration of the antigen bovine albumin was 0.17 γ /ml in a 2 per cent normal rabbit serum. An interesting fact emerges from these experiments: the differences $\Delta\text{\AA}$ in the thickness adsorbed from an antiserum on two areas of a slide, one having been treated with a 2 per cent normal rabbit serum containing bovine albumin and the other with a 2 per cent normal rabbit serum (carrier) were of the same order of magnitude, even when the concentrations of antigen were changed by a ratio of 1-1000. Similar results were obtained with growth hormones as antigens, suggesting that this might be a general phenomenon.

*Determination of growth hormones**

As mentioned earlier, reproducibility of the results obtained by IEA is greatly facilitated if the metalized slides are coated with five layers of Ba stearate before performing the test. All the results reported in Table 3 were obtained with such prepared chromium plated slides. Tests were performed with a very pure

Table 3. Difference in thickness in Ångström units ($\Delta\text{\AA}$) between the layers adsorbed from an antihuman growth hormone rabbit serum on two slides, one coated with growth hormone and carrier and the other with carrier only

HGH m μ /ml	0.0	0.1	0.2	0.34	0.5	1.0	5.0	16	55	200	1000	20,000 in water only
$\Delta\text{\AA}$	0	0	3	10-15	12	15	20	18	18	18	19	21

human growth hormone preparation provided by Dr. Hao Chia Chen of the Rockefeller University. The growth hormone had an activity of 2.1 USP units/mg and a prolactin activity of 7.4 IU/mg. Human growth hormone was dissolved in a carrier consisting of a 2 per cent normal human serum buffered with veronal pH 7.7, 0.0015 M. The slides were positively charged for the adsorption of the hormone and of the immune serum. The adsorption time was half a minute for the antigen and one minute for the antiserum. The intensity of the current was kept at 300 μ A for both adsorptions. The first row of Table 3 gives the concentrations of growth hormone expressed in m μ g/ml. The figures in the second row $\Delta\text{\AA}$, represent in angströms the difference in thickness of the layers adsorbed from the immune serum on two slides, one coated with growth hormone plus carrier and the other with carrier only. The thickness adsorbed on slides coated with carrier was of the order of 18-25 Å, the variations being due to the time elapsed after the coating of the slide with layers of stearate. This necessitated running a control after each experiment, as was done in the case of bovine albumin. For an assay of growth hormone, the sample was diluted with a carrier

*All experiments concerning growth hormone were carried out in collaboration with Dr. E. H. Thiele of the Merck Institute for Therapeutic Research, Rahway, New Jersey.

until the value $\Delta\bar{A}$ dropped sharply below 17, corresponding to a concentration of $\approx 0.3 \text{ m}\mu\text{g/ml}$. Consequently, the level of growth hormone units in the sample was simply $0.3 \times$ dilution. The concentrations of growth hormone in two sera, one from an acromegalic and the other from a hypopituitary subject were found to be $12 \text{ m}\mu\text{g/ml}$ and $0.8 \text{ m}\mu\text{g/ml}$ respectively.

DISCUSSION

It is a remarkable fact that such minute quantities of growth hormone can be estimated by this method under the experimental conditions utilized. At the limit of sensitivity, there are in the solution roughly ten million molecules of serum proteins to one molecule of growth hormone.

It is easily calculated that the average thickness of a layer of growth hormone adsorbed from 0.5 ml of a solution containing $5 \times 10^{-10} \text{ g/ml}$ of hormone, cannot be greater than $0.03 \text{ \AA}$ which is the average thickness attained if all the molecules of growth hormone present in 0.5 ml had been adsorbed on 0.6 cm^2 which is the immersed area of the slide. The diameter of a growth hormone molecule is $\approx 35 \text{ \AA}$, assuming a molecular weight of 22,000. The molecules are probably adsorbed in their native state. This means that at most 0.1 per cent of the area of the slide is occupied by the adsorbed growth hormone molecules. Practically the whole area is coated with molecules adsorbed from the carrier. The layer adsorbed from an immune serum on a slide so sparsely populated with growth hormone molecules is $20 \text{ \AA}$ thicker than the layer adsorbed from the same immune serum on a slide coated with the carrier only. The average 'diameter' of a rabbit antibody molecule is of the order of $65 \text{ \AA}$. It follows that a minimum of 200 molecules of antibodies and probably many more have been immobilized per molecule of antigen. When the concentration in growth hormone of the solution used for the electroadsorption is $5 \times 10^{-9} \text{ g/ml}$, it can be estimated that the average distance between the adsorbed growth hormone molecules is at least of the order of $1000 \text{ \AA}$. The area occupied by 200 molecules of antibodies is approximately $\approx 2 \times 10^6 \text{ \AA}^2$, which is of the same order of magnitude as the free area between each molecule of antigen ($1000^2 = 10^6 \text{ \AA}^2$). In other words, the whole area of the slide is already covered with one layer of antibody molecules, one molecule thick (saturation) when the average distance between the underlying growth hormone molecule is $\approx 1000 \text{ \AA}$. This would explain why there is no further increase in the thickness of the layer of antibodies when the concentration of growth hormone in the solution is larger than $5 \text{ m}\mu\text{g/ml}$. Experiments performed without a current with concentrated solutions of pure hormone have shown that the maximum thickness adsorbed from an immune serum was $\approx 35 \text{ \AA}$ corresponding to less than a single molecular layer.

It is difficult of course to visualize the mechanism which permits the immobilization of such a large number of antibody molecules by a few adsorbed molecules of growth hormone. One should await further experimental results before discussing profitably the complex interactions involved.

A large ratio of antibody to antigen molecules is also found for other systems. For instance, a layer of antibodies up to $500\text{--}600 \text{ \AA}$ thick can be specifically adsorbed on an adsorbed layer of polysaccharide type III pneumococcus $2 \text{ \AA}$ thick, no current being used for the adsorption[11]. Thus a very large number of

antibody molecules can be specifically immobilized on a solid surface by the presence of a single antigen molecule already adsorbed on this surface. It is this surprising fact which renders the IEA assay sensitive enough to detect the presence of $\approx 2 \cdot 10^{-10}$ g/ml of growth hormone.

The influence of the anchorage of stearate layers on the immunological reaction

As mentioned above, it was found that very consistent results were obtained if the slides were first coated with five monolayers of Ba stearate prior to the adsorption of the antigen. This seems true not only for the antigen human growth hormone system, but also for serum bovine albumin and its antiserum. If three monolayers were substituted for five layers entirely negative results were obtained, the thickness adsorbed from the immune serum on the slide coated with carrier plus antigen being the same or even smaller than that adsorbed from the carrier coated slide. In the system human growth hormone, with the increase in the number of layers of stearate in the anchorage, positive results were obtained more often when the anchorage consisted of nine layers rather than seven or thirteen. It was important to use the slides within 4–6 hr after the deposition of the stearate layers. No positive results could be obtained with slides coated 20 hr before they were used for an IEA test. In other words the thickness observed after the electroadsorption of the immune serum was the same on two slides, one covered with antigen plus carrier and the other with carrier only. However, if the slides were kept in a test tube in a water vapor saturated atmosphere they could be used to detect immunological reactions just as well as freshly prepared slides, at least 24 hr after their preparation. Slides freshly coated with five layers of stearate and heated for 5 min at 50°C immediately lose their property for identification of immunological reactions. The same thing happens if the slides are placed under vacuum for 10 min.

Slides coated with five stearate layers are perfectly dry coming out of the trough. After heating a slide coated with five layers at 50°C for 5 min one can observe a small decrease in optical thickness of about 2 Å. If the heating is done after the layers of antigen have been adsorbed, a higher temperature (60° for 10 min) is required to prevent the slide from differentiating the specific interaction from the control. A few experiments were conducted in which a film of formvar 130 Å thick was substituted for the five layers of Ba stearate. Such prepared slides were perfectly adequate to detect immunological reactions in the system human growth hormone and the corresponding rabbit immune serum. In this case also, the presence of a small amount of water in the film seemed a necessary condition for carrying out successfully the IEA test. The fact that the electroadsorptions of both antigen and antibodies are carried out in water makes these results even more striking. Once a slide coated with five layers of stearate has been aged or dehydrated, it cannot regain its property for detecting an immunological reaction even if the slide is kept for a long time in a water vapor saturated atmosphere. The phenomenon appears irreversible. These results concerning the great influence of the number of inert layers in the anchorage on the subsequent immunological reaction occurring on the top layer, show the subtlety of the mechanisms involved. As mentioned earlier, an anchorage of stearate layers is not a *sine qua non* condition for carrying out

successfully an IEA test. If the metallic surface is adequate, no anchorage is necessary. However, if three layers of stearate are deposited on an adequate metallic surface, the slide becomes unable to identify an immunological reaction. The number of layers in the anchorage is of prime importance. A periodicity in the detection of the immunological reaction appears to occur as a function of the number of subjacent layers of stearate. The aging of the slides, which renders them insensitive for the identification of an immunological reaction, results from a loss of water, with possibly a disruption in the orderly arrangement of the layers.

When considering all these facts, it appears that the bringing together, with subsequent interaction, of the active groups of the adsorbed antigen and antibodies molecules involves indirectly all the layers of the anchorage. Cooperative phenomena apparently take place which will or will not allow the specific immunological adsorption to occur on the top layer of the anchorage. The occurrence of the specific immunological adsorption appears to be controlled by a wave-like mechanism. The periodic influence of the number of underlying stearate layers on the immunological reaction is in harmony with facts already reported years ago concerning the rate of diffusion of trypsin molecules through a thin membrane of formvar protecting a few monolayers of protein[12]. The protein layers were deposited on an anchorage of one, or three, or five, or seven, etc. monolayers of Ba stearate. It was observed that the rate of forced diffusion of trypsin through the Formvar membrane was a periodic function of the number of layers of stearate located under the protein layers. The rate of diffusion was the slowest by a large factor when the protein layers were placed on top of three or eleven monolayers of fatty acid. The position of the minimum in the rate of diffusion as a function of the number of stearate layers could be somewhat displaced depending on the presence of inert membranes placed under the anchorage. For these experiments no electric current was used, contrary to those reported in this article.

CONCLUSIONS

The fundamentals involved in the IEA assay have been discussed and empirical rules based on experimental results have been proposed which will permit one to conduct successfully an IEA test. Without a doubt the method has not yet reached its final stage, and many improvements are bound to be made as long as the subtle phenomena which do control these specific adsorptions are not better understood. In the meantime, the findings reported here should prevent other laboratories interested in this new method from going through the difficulties encountered when we were first confronted with this new approach.

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